

Post-detection related developments, and SETI in Society news.

An update for the IAA SETI Permanent Committee, 26 Oct 2021, at IAC Dubai

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Overview of what will be covered:

- Current status: since KD and Les Tennen last provided a post-detection-related update in 2019
- Papers on themes related to Post-detection in 2021 and 2020 IACs
- Some relevant recent and upcoming activities of SETI Permanent Committee members
- Some recent work of relevance in the larger research community
- Some topics of general relevance: UAPs, COVID19 information research, scientists in society, diversification of potential detection scenarios
- Member reports: Daniela DePaulis, Stelio Montebugnoli, Andrea Melis, on A Sign In Space, and John Elliott on UK SETI Research Network activities related to post-detection
- BLC1 (Breakthrough Listen Candidate 1) update: still not aliens, but informative: a few comments then opportunity for Q&A with BL team members
- General Q&A

General Status

- We have many members interested in post-detection, but we aren't currently formally organized into a working group with a specific remit. To be revisited.
- The last update in 2019 suggested some specific ways forward.
- COVID interruptions have been major, of course, but hopefully there will be some post-detection workshops etc. in the next couple of years.
- Meanwhile, considerable research relevant to PD does continue
- It would be beneficial for non-astronomers on the committee to report back to the committee regularly concerning relevant developments in other fields

1. Papers related to Post-detection: 2021 and 2020 IACs

2021: <https://iafastro.directory/iac/browse/IAC-21/A4/2/> with presenter name (follow links for full author teams)

- Steps to follow after establishing contact with extra-terrestrials, Mr. Aditya Balasubramaniam
- The need for a worldwide and international SETI Journal, Dr. Claudio Maccone
- Potential Technologies Through SETI and Global Response Plan for First Contact, Dr. Ugur Guven
- “The readiness is all” - Understanding Post-Detection Dynamics through Live Action Role Play, Ms. Kate Genevieve
- On the necessary aspect of METI, Dr. Carl DeVito
- Astropolitics and the Implications of Belonging to an Extraterrestrial Polity, Mr. Mclee Kerolle
- How law and ethics would need to adapt to the discovery of life, Mr. Jordi Sandalinas
- Benedict XVI and SETI, Dr. Paolo Musso

1. Papers related to Post-detection: 2021 and 2020 IACs

•**2020:** <https://iafastro.directory/iac/browse/IAC-20/A4/2/> , with presenter name (follow links for full author teams)

- The Duty Under the Space Law Treaty Regime to Disclose or Report an Encounter With an Extra Terrestrial Intelligence, Mr. George Anthony Long
- Protocols to be Implemented in Extraterrestrial Contact and SETI Security Protocols, Dr. Ugur Guven
- The InCosmiCon Research Center and its activities in the field of SETI, Big History and interculturality, Dr. Paolo Musso

2. Some relevant recent and upcoming activities of SETI Permanent Committee members

- N.b. This compilation is not comprehensive. Also see updates below (sections 5 and 6)
- **Continued public outreach by SETI Institute and Breakthrough Listen**
- **Awaiting results of Astro2020 Decadal Survey:** released Nov 4 www.nationalacademies.org/our-work/decadal-survey-on-astronomy-and-astrophysics-2020-astro2020 Members submitted multiple papers concerning technosignatures including one on the need for sustained post-detection research funding (Denning & Dick). <https://baas.aas.org/astro2020-apc> , <https://baas.aas.org/astro2020-science>
- **PSETI (Penn State) activities including:**
 - PSETI 2022 symposium... perhaps there will be a postdetection session. <https://sites.psu.edu/setisymposium2021/>
 - SETI Bibliography Project <https://arxiv.org/abs/2107.02887>
 - SETI News <https://seti.news/about/>
 - SETI Seminars
 - The Assembly of the Order of the Octopus student-run, with multiple contributions touching on post-detection, Archive at: <https://zenodo.org/communities/octopusassembly2021/>

2. Some relevant recent and upcoming activities of SETI Permanent Committee members, cont'd

- N.b. This compilation is not comprehensive.
- **Technosearch:** <https://technosearch.seti.org/> For anyone who missed its launch in 2019: obviously a wonderful resource. In the present context: it is useful for thinking about different types of detections. How is this going, i.e. are papers being submitted? It could also be a useful model for collecting post-detection-related research.
- **1I/'Oumuamua:** first known interstellar object, observed Oct 2017 for a couple of months (then out of range), papers have continued, most arguing natural origin, (e.g. Desch & Jackson, Mar 2021), but also the contrary view e.g. Avi Loeb's Jan 2021 book *Extraterrestrial: The First Sign of Intelligent Life Beyond Earth*. Some controversy, but also features instructive regarding PD: inaccessible for further observation, lag time for analysis, major difference of opinions between scientists. Also reinforces argument for solar system SETI (objects).
- **The Galileo Project for the Systematic Scientific Search for Evidence of Extraterrestrial Technological Artifacts**, launched July 2021, led by A. Loeb, with a considerable team. <https://projects.iq.harvard.edu/galileo>
"The goal of the Galileo Project is to bring the search for extraterrestrial technological signatures of Extraterrestrial Technological Civilizations (ETCs) from accidental or anecdotal observations and legends to the mainstream of transparent, validated and systematic scientific research. This ground-based project is complementary to traditional SETI, in that it searches for physical objects, and not electromagnetic signals, associated with extraterrestrial technological equipment."
- **Watch for papers and sessions at SSOCIA 2022 and at AbSciCon 2022** (paper abstracts still being accepted for both)

3. Some recent work of relevance in the larger research community

- A list of some relevant recent papers is included in the report to the Committee. There is ongoing interest from many corners concerning post-detection matters, in journals including JBIS, Space Policy, IJ Astrobiology, Acta Astronautica, Am J International Law, Astrobiology, etc. Topics include: public interest in extraterrestrials; political considerations and possible political problems after a detection; first contact protocols; legal questions about UN involvement in contact scenarios; metalaw questions concerning relationships with extraterrestrial societies; public opinion concerning hypothetical future responses to ETI, etc.
- There have also been some great talks and workshops which are available online: “Sanctuaries and the Study of Cetacean Communication”, “SETI: Cognition And Communication In Extraterrestrial Intelligence”, Technosignatures seminar series at Goddard, papers at SSOCIA 2020.

4. Some topics of general relevance

- a) **Increasing and changing discussion around UAPs** -- an important part of the landscape. The release of this American report in June 2021 garnered much interest and much coverage. www.dni.gov/files/ODNI/documents/assessments/Preliminary-Assessment-UAP-20210625.pdf
- b) **The COVID-19 pandemic: applicable lessons about mis/dis/information...** A great deal of applied research concerning the spread of misinformation and disinformation – building upon a substantial body of work that already existed concerning, in particular, the various effects of social media.
- c) **Scientists in society: lessons from climate and from COVID.** Recent surveys demonstrate that an alarming number of scientists communicating with the public about divisive topics – like climate change and COVID-19) – have experienced harassment, online abuse, hostility, threats. e.g. www.nature.com/articles/d41586-021-02741-x Social media safety for researchers is a concern.
- d) **Ongoing Diversification of Potential Detection Scenarios.** Significant new projects and instruments mean more and different potential detection scenarios. Significant new players in SETI: in particular, the rising profile of FAST and SETI work in China. Dr Di Li gives a great update here about FAST activities including SETI, Feb 2021: www.youtube.com/watch?v=GOcJ14HA4Xg in the IAF session on “Breakthrough Listen and International SETI Collaborations”. Additionally, new instruments either for technosigs or for exoplanets and biosigs (e.g. JWST, Dec launch), could all result in different detection scenarios, with different implications for public discussion.

5. Member reports

**a) A Sign in Space, by Daniela DePaulis, Stelio Montebugnoli,
Andrea Melis**

Full presentation with narration at:

<https://drive.google.com/file/d/16fHGWZHImc8oik7OWXQJQ6LeR01Z8Ll/view>

**(b) John Elliott, UK SETI Research Network Post-Detection
Activities: JE will speak to this (slides below)**



UK SETI RESEARCH NETWORK POST-DETECTION ACTIVITIES 2019 -21

"WORKING NATIONALLY FOR INTERNATIONAL EFFECT."

DR JOHN ELLIOTT
CHAIR UK SETI RESEARCH NETWORK (UKSRN)
HONORARY RESEARCH FELLOW
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Current UKSRN PD research

- ▶ Communication Theory
- ▶ Social Media /Societal impact analytics & strategies
- ▶ Protocols
- ▶ Space Law
- ▶ Metalaw / Legal impacts on Human Civilisation
- ▶ METI implications
- ▶ LARP (Live Action Role Play) – paper at this conference:
 - ▶ *THE READINESS IS ALL” - UNDERSTANDING POST-DETECTION DYNAMICS THROUGH LIVE ACTION ROLE PLAY* Genevieve, K. Kershenbaum, A. Elliott, J. & Catlow, R.
- ▶ Signal categorisation, decoding & analytics
- ▶ Royal Society - From Afar Science exhibition <https://www.fromafar.world/> - ongoing survey on Contact
- ▶ Theological implications of Contact
- ▶



Who are we?

Membership includes (far from an exhaustive list – apologies for those missed of on this one slide):

- Patron – Professor Lord Martin Rees – University of Cambridge, Astronomer Royal
- Dr John Elliott (Chair) - University of Manchester – Computational linguistics, language universals, Xenolinguistics, decipherment, global impact strategies and information dissemination.
- Professor Michael Garrett - University of Manchester & Jodrell Bank Observatory - detection of techno-signatures & searches using wide-field and interferometry techniques.
- Dr Martin Dominik - University of St Andrews - gravitational bending of light for studying the demographics of planets
- Dr Stephen Baxter - British Interplanetary Society & Sci-Fi Writer - Post-detection policy, Metalaw, METI.
- Dr William Edmondson - University of Birmingham – Cognitive Science & SETI (Search)
- Professor Michael Bolander - University of Durham – Law (International Judge, The Hague)
- Professor Christopher Newman - Northumbria University – Space Law & Policy
- Revd Professor David Wilkinson - University of Durham - Religious implications & challenges of SETI
- Dr Arik Kershenbaum – University of Cambridge – Zoologist: recent book The Zoologist's Guide to the Galaxy
- Professor Ian Crawford – Birkbeck College, University of London - Planetary Science and Astrobiology
- Dr Anders Sandberg – University of Oxford – Futures of Humanity, Physical eschatology
- Professor João Pedro de Magalhães - University of Liverpool - Bioinformatics; Evolution; Genetics; Machine Learning
- Dr Klara Anna Capova – University of Durham & European Space Agency - sociocultural aspects of SETI and METI activities & Anthropology.
- Kate Genevieve – University of Sussex - Multi-sensory communication, post-detection, LARP.
- Dr Stephen Webb - University of Portsmouth - Approaches to the Fermi paradox / Great Filter, Possible role of compact objects in SETI
- A full list will be available on our new web site (to be available soon), which will list all members, their expertise and main publications.

One example of our UKSRN PD recent activity

Professor Michael Bohlander | Durham University Law School Preparing a baseline for human negotiations with ETI (“Unilateral Metalaw”)

Chair in Comparative and International Criminal Law | International Criminal Judge

Member of UKSRN

Member of International Institute of Space Law

Member of British Interplanetary Society

[l-bohlander/](https://www.durham.ac.uk/staff/michael-bohlander/)

www.durham.ac.uk/staff/michael-bohlander/

- ▶ Metalaw – What is it good for?, (2021) 188 Acta Astronautica, 400 – 404- doi: <https://doi.org/10.1016/j.actaastro.2021.06.039>
- ▶ Joining the “Galactic Club”: What Price Admission? – A hypothetical case study of the impact of human rights on a future accession of humanity to interstellar civilisation networks, (2021) 132 Futures, 102801 – doi: <https://doi.org/10.1016/j.futures.2021.102801>
- ▶ Current monograph research project: “Conflict with Extraterrestrial Intelligence and Human Law – A speculative study of the impact of interspecies conflict on humanitarian and human rights law” (Submission mid- to late 2022)

Main research questions:

- ▶ Legal aspects of human reaction to contact with ETI (including direct hostile contact)
- ▶ Legally binding regulation of post-detection regimes
- ▶ The need for a human cosmic species awareness



Towards a Comprehensive SETI Post-Detection Environment: Creating a **Post-Detection Hub for International Collaboration and Coordination.**

1. A problem many of us recognise: post-detection is a matter of considerable significance for SETI, but has often suffered from inadequate coordination, insufficient research, and a lack of funding.

UKSRN's proposed contribution to the solution: a Post-detection Hub which draws interested members of the SETI community together to work on topics ranging from message decipherment and data analytics to development of regulatory protocols, space law and societal impact strategies.

2. We are looking to include organising symposia, seminars and conferences, presentations, publications, funding applications and drafting proposals for, as well as meetings with, external, and especially with domestic and international political and governmental, stakeholders, and activities of a similar nature.



Also in the Pipeline

- ▶ A focused, dedicated international conference on all matters pertaining to Post Detection.
- ▶ We'll be reaching out to everyone soon, to make this a truly International endeavour. Please contact me, if you are interested. setijon@aol.com
- ▶ Watch this space.



6. BLC1 (Breakthrough Listen Candidate 1) update: still not aliens, but informative

- For the full story and papers with full published analysis, see <https://seti.berkeley.edu/blc1>. n.b. New signal verification 'toolkit'. BL team members are here, if you have questions.
- From post-detection perspective:
 - thanks are due to many excellent journalists and science writers
 - kudos: a rapid response from the BL team and other astronomers was effective in getting accurate information out
 - this case reiterates the challenges for SETI PD, including premature and ill-timed leaks, the fact that this science takes time (i.e. prolonged analysis time, and particularly if follow-up observations seem warranted), and explaining complex technical aspects to non-experts, etc.
 - this is a very instructive case for us all
 - if the signal had passed more of the tests, or had seemed complex, there could have been more room for speculation, more for everyone to handle
 - a refinement to Rio2 may be worth pursuing -- or an alternative

End. Questions?

**Below are slides from the 2019
IAA SETI Committee PostDet
update, in case they are needed
for discussion**

Composition of IAA SETI Committee in mid-2018 (~80)

<https://iaaseti.org/en/members/>

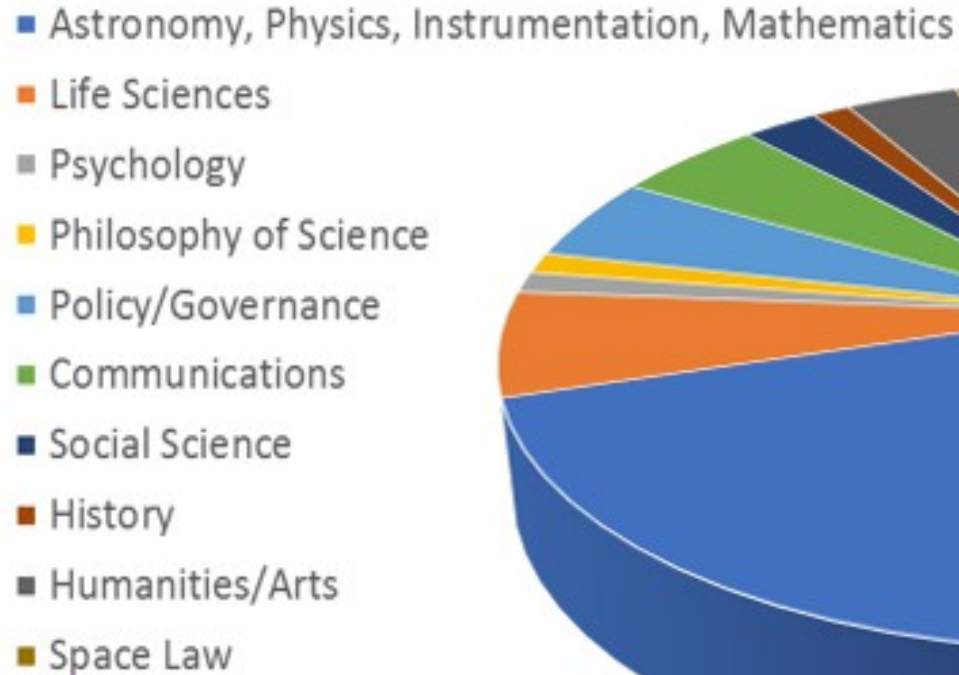
For reference:

Membership in this committee has fluctuated over the years, but it rose sharply several years ago.

- In 2002, 37 members (~ half had expertise other than astronomy/physics)
- In 2009, 43 members
- In 2014, 40 members
- In 2019, over 80 members

Also, as of mid-2018:

- female proportion of committee is < 20%
 - ~ 50% of committee is American
- A survey would tell us more (age etc)



Several suggestions flow from that reality of committee composition:

- Recognize that we do not have all the necessary expertise in-house for social & communications topics – we must draw on external expertise and research, but we should also bring more diverse experts in
- Hold a series of virtual meetings which are small enough that people can actually talk to each other
- Gather some information from committee members about interests/skills
- We need to recognize as well that some members cannot travel to IAC meetings and can only ever participate remotely.
- Given the attempt to integrate SETI/technosignatures and astrobiology, i.e. technosignatures are biosignatures... perhaps we need more astrobio.

Protocols and Post-Detection... reloading

The Status:

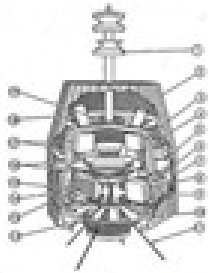
- 2010 Principles (revision of 1989) are already outdated
 - Post-detection Committee (*which is a part of those Principles*) hasn't existed for a few years and still needs to be reimagined
 - Rio (*the use of Rio or something like it, is part of the Principles*) has been improved into Rio 2.0 with significant strengths, but can be further adjusted, and/or a complementary strategy for public communications could be developed
 - The Principles themselves may need an update but as Tennen and Forgan concluded in their papers over the last two years, some social research would help us figure out the path. But this isn't easy.
 - This is one reason why Denning et al 2019 (Berea, Capova, Forgan, Korpela, Oman-Reagan, Tarter) suggested research roadmapping to figure out *what* we need to know, and how we can get that information. Thanks to Bill Diamond/SETI Inst for agreement in principle to move forward with this. (details coming)
 - Many of the big challenges stem from huge changes in the communication landscape.

Evolution of communications systems since modern SETI began

1960s



1970s

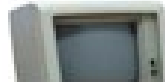


1980s



@

1990s



2000s



2010s



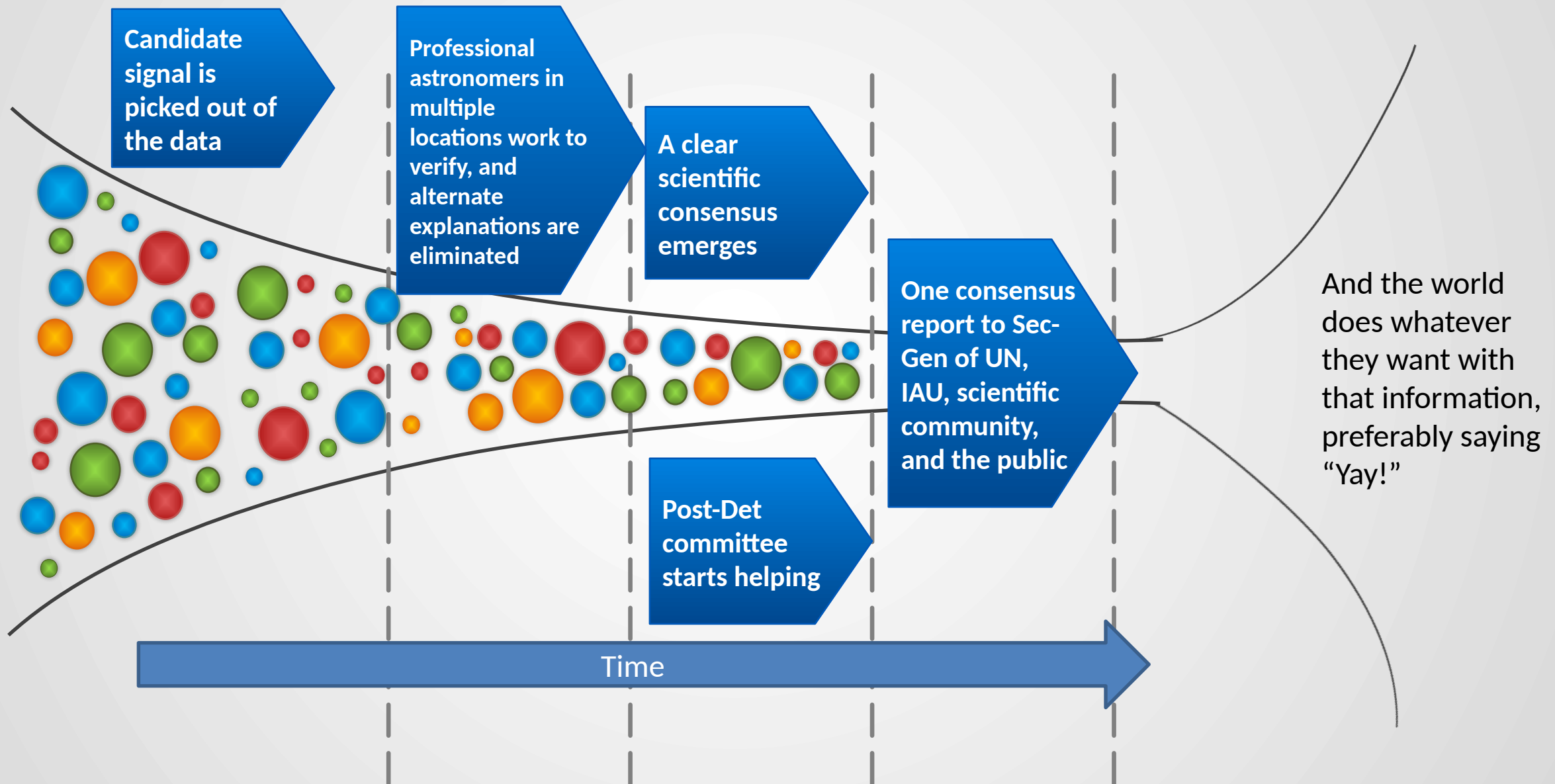
2020s



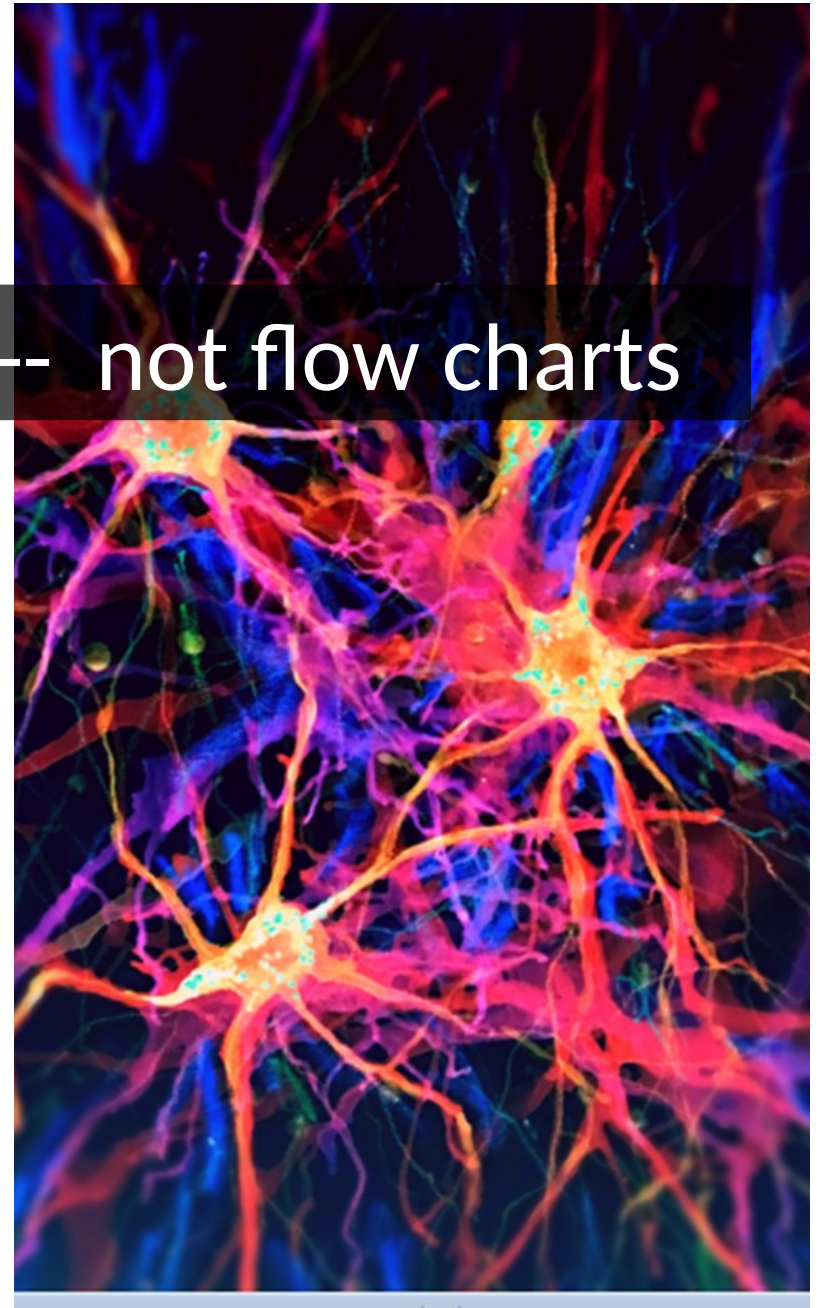
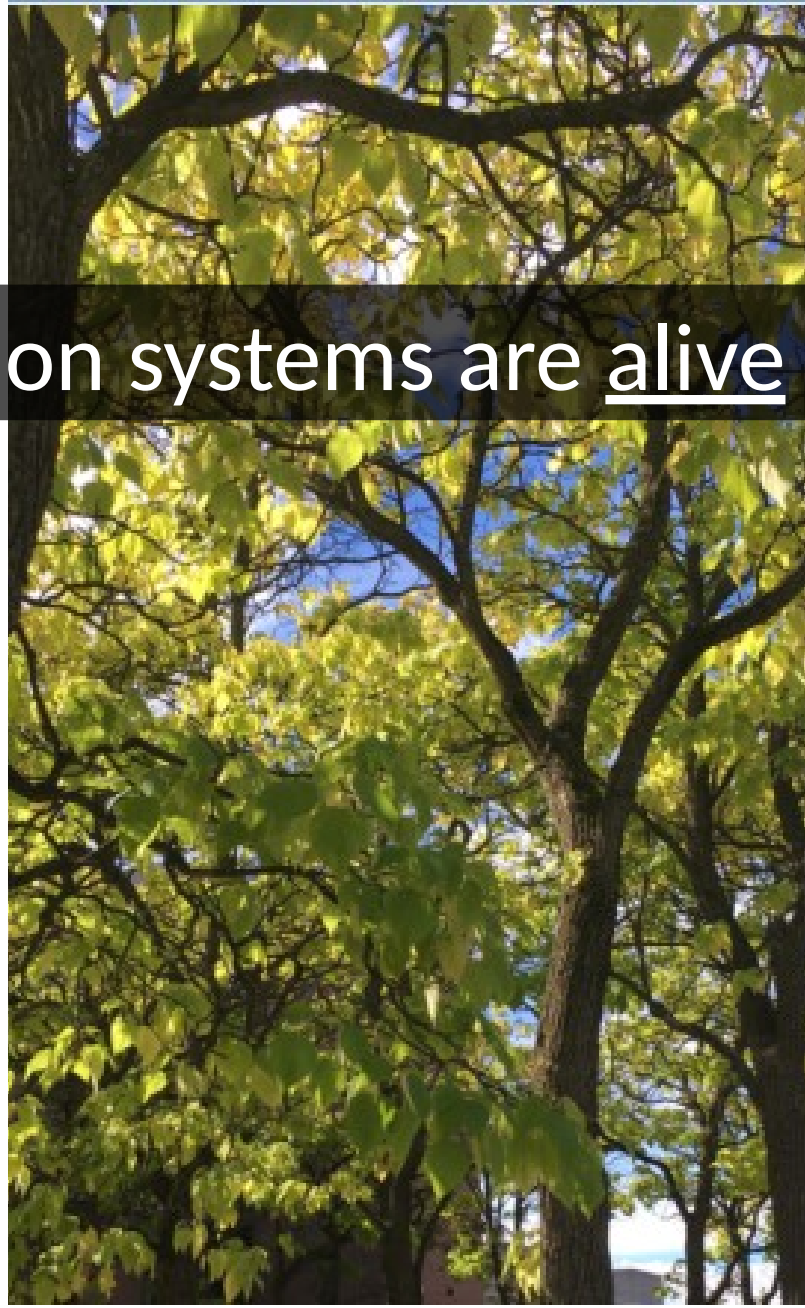
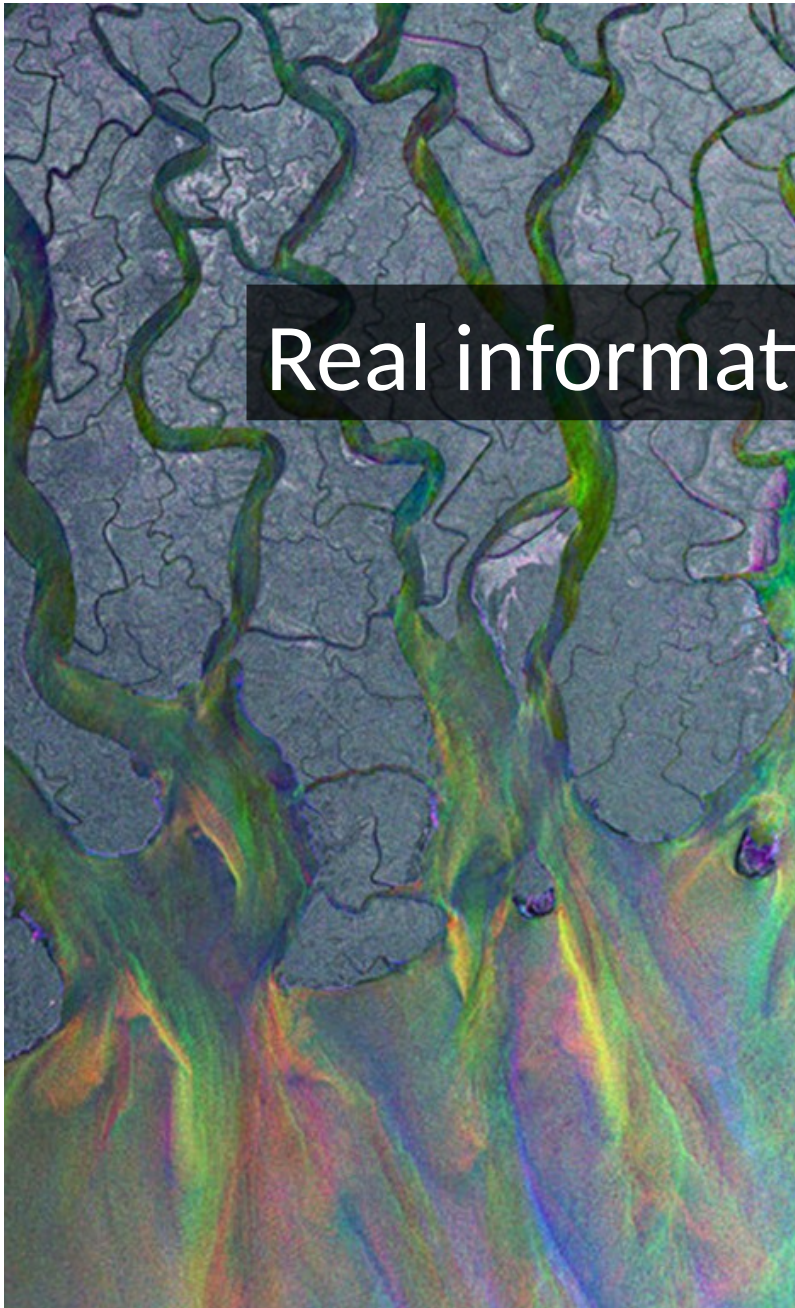
**Now: peer-to-peer, unmediated, fast, and wide:
Increase in direct communication between scientists
and scientists, scientists and public, faster
information transmission, etc...
Has implications for how we handle
announcements, debates, discussions**



Idealized process implicit in original Principles



Real information systems are alive -- not flow charts



Approaches and bodies of research that can help (there are more) (from Denning et al 2019)

- Information flow studies (information seeking behavior, including radio, TV, posters, word of mouth, text)
- Social media studies (including negativity bias, mis/disinformation, profit)
- Effective brief communication (e.g. emergency warning systems)
- Not just words: also images
- Mixed methods: interviews, focus groups, analogy, careful survey design
- Legal and Economic questions

so.... how do we move forward?

ultimately, by going back to the spirit of the Principles, rather than the letter

- Original intentions (approximately):

- encourage researchers in different countries to share, by *mutually* promising to be forthcoming
- to do responsible government-sponsored science
- be transparent with the public
- supply reliable, verified information to the world
- be worthy of trust
- Assure everyone that SETI researchers would not exceed scientific remit, i.e. would stop short of exopolitics, i.e. let world governments step in, if contact actually is made

And now we must ask:

- what does scientific responsibility look like *now*?
- what are the situations to be avoided now?
- what would a discovery look like *now*?
- what is in our hands, and what is out of our hands, *now*?
- how have our publics changed?

Suggested directions (these interconnect)

Aim for Principles AND Codes of Conduct AND Best Practices

Timeless principles

Voluntary
codes/agreements
among all levels of
research (incl
students, citizens)

Regularly updated,
research-based best
practices suggestions

Research roadmap with broad scope

Social / communication
research topics such as
these above,

AND

Ethics

Studies of evolution of
intelligence &
communication

Studies of language /
decipherment

Urgent priority areas include:

Concerns of involved
scientists

Information seeking and
sharing behavior

Areas of public concern

Best practices for social
media use for
researchers

So, recommendations:

- 1) An informal, small interdisciplinary Committee working group for one year, which will rethink and recommend which subcomms we will need.
- 2) Hold a series of small Zoom meetings for Committee members to find out interests, expertises, and skills
- 3) Small groups to produce some draft Codes of Conduct – to be brought back to this committee next year
- 4) Small groups to produce some draft Best Practices – e.g. Social Media Safety – to be brought back to this committee within a year
- 5) Move ahead with Research Roadmapping for social research related to technosig discovery process and public communications. (KD will begin that process, with SI.)
- 6) And?